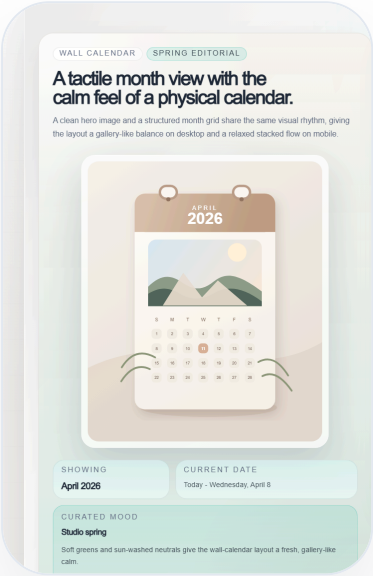
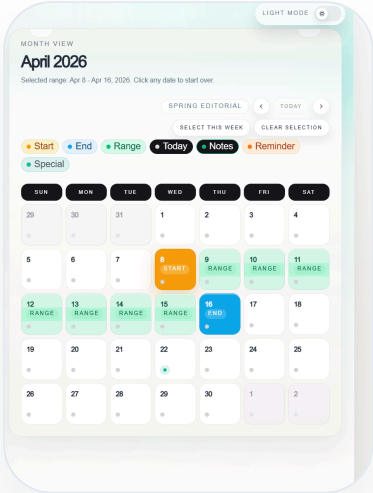
FRONTEND ENGINEERING CHALLENGE REPORT

Interactive Wall Calendar Component

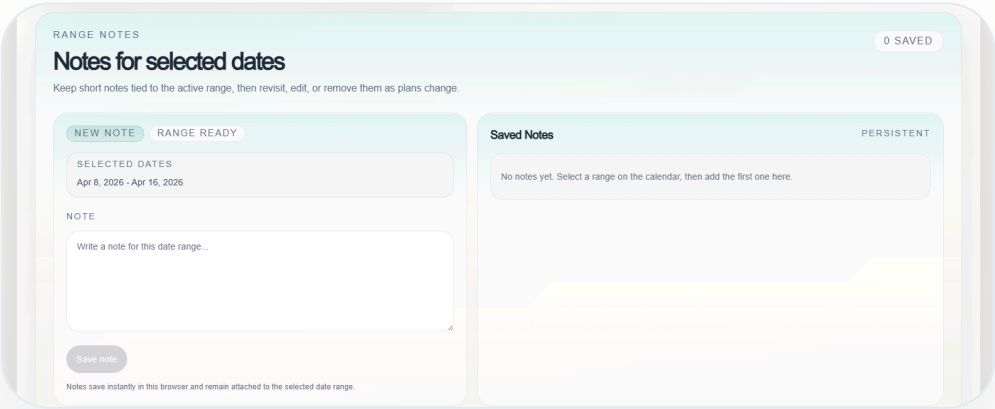
An enhanced version of the original wall calendar challenge, expanded with production-minded UX, advanced interactivity, accessibility support, responsive behavior, persistent notes, reminders, and premium interface polish.



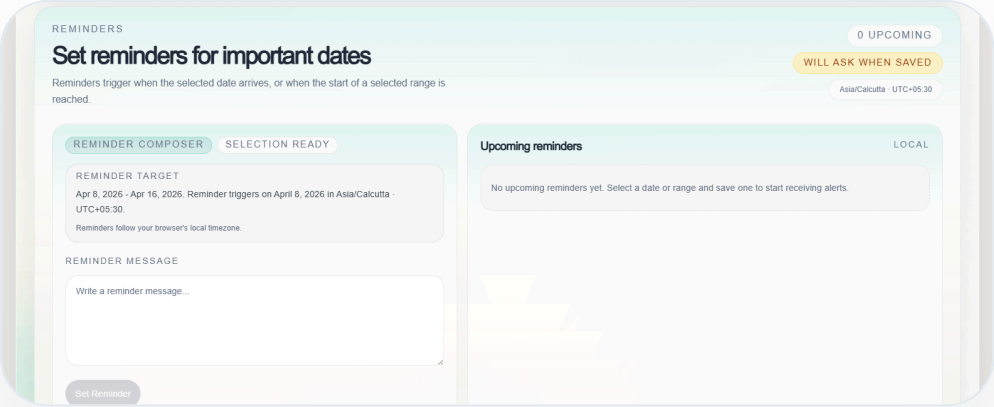
Hero composition with wall-calendar styling, editorial framing, and seasonal accenting.



Month view with a highlighted date range, quick actions, and important-date markers.



Notes workflow connected to the active range with clear save and empty-state handling.



Reminder workflow with timezone-aware scheduling, permission state, and upcoming reminder visibility.

Project Summary

This project began as a UI-focused calendar brief and was intentionally pushed into a richer product-style experience. The result is a wall-calendar-inspired month view that supports interaction patterns users expect from modern planning tools while keeping the design clean, balanced, and recruiter-ready.

Focus: UX quality, motion restraint, accessibility, responsive layout design, and real-world usability beyond the original challenge scope.

Implemented Features

Calendar and Range Selection

- Wall calendar layout with hero image and month-view calendar grid
- Current month display with 7-column day grid and current-day highlight
- Click-based and drag-to-select date range interaction

Notes and Reminders

- Notes system with add, edit, delete, and localStorage persistence
- Reminder creation for a selected date or date range
- Reminder message support and upcoming reminders list

- Clear start, end, and in-range highlighting
- Quick actions for Today, Clear Selection, Select This Week, and month navigation
- Smooth month transition animation with slide and fade behavior

- Browser notifications with in-app alert fallback
- Timezone-aware reminder display and scheduling
- Calendar indicators for notes and reminders

UX, Visual Design, and Motion

- Responsive design for desktop, tablet, and mobile layouts
- Light and dark mode with saved preference
- Dynamic accent colors influenced by the hero image
- Framer Motion transitions for load, selection, panels, and month changes
- Sticky desktop notes workflow and smoother panel hierarchy
- Hover states, focus states, toasts, and premium feedback styling

Accessibility and Production Readiness

- Keyboard date navigation with arrow keys, Enter, and Space support
- Visible focus treatment and screen-reader-friendly labels
- High-contrast styling and dark mode readability improvements
- Edge case handling for reversed ranges and same-day selections
- Validation for empty notes, incomplete reminders, and permission denial
- Client-side persistence with maintainable TypeScript-based architecture

Design Decisions

Why the wall calendar concept was chosen

The wall calendar format creates a stronger visual identity than a typical date picker. It feels more editorial, more memorable in a portfolio, and still keeps the month view easy to understand at first glance.

Why localStorage was used instead of a backend

The challenge is frontend-first, so localStorage keeps the project self-contained, easy to review, and functional without requiring authentication, APIs, or cloud infrastructure.

How UX was handled

The interface was refined around real interaction quality: drag selection, quick actions, mobile touch targets, visible feedback, accessibility support, and clear fallbacks when browser permissions are limited.

Tech Stack

Next.js 16 App Router | TypeScript | Tailwind CSS | Framer Motion | date-fns | localStorage

Future Improvements

- Backend integration for cloud persistence and cross-device sync
- User authentication and personalized calendars
- Cloud-based reminders with reliable background delivery
- Multi-user collaboration and shared planning features

Prepared as an enhanced submission package to show strong frontend engineering judgment, not only feature completion.